

Review

Agricultural Robotics: A Technical Review Addressing Challenges in Sustainable Crop Production

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Abstract: The adoption of agricultural robots is revolutionizing the agricultural sector, offering innovative solutions to optimize production and reduce environmental impact. This review examines the main functions and applications of agricultural robots, with a focus on the crops handled and the technologies employed. The study analyzes the current state of the art regarding the market trend of agricultural robots used in field and greenhouse operations. Several solutions are emerging, some already implemented and others still in the prototype or project stage. These solutions are beginning to spread, though they may still seem far from widespread field application, particularly given the peculiarities and heterogeneity of the global agricultural landscape. In the face of the many benefits associated with the use of agricultural robots, even today some technical bottlenecks and costs limit their widespread use by farmers. The review provides a fairly comprehensive and up-to-date overview of current trends in agricultural automation, suggesting new areas of research to improve the efficiency and adaptability of robotic systems to different types of crops and environments.

Keywords: autonomous ground vehicles; automation; artificial intelligence; environmental sustainability; crop management

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1. Introduction

Precision agriculture, which encompasses the increasing use of information technology, sensors, autonomous vehicles, data analysis, predictive modelling and other digital solutions for agricultural activities, is often seen as a promising solution to improve food security, reduce water consumption, decrease fertilizer and pesticide use, and increase farm profitability. However, despite these potential benefits, the adoption rate of advanced agricultural technologies remains low and varies widely depending on the specific technology and geographical region considered. At the same time the economic benefits must be compared with the implications and challenges involved, such as the initial investment required and the specialized technical skills for operation and maintenance [1,2].

Governments globally aim to increase productivity and improve farmers' livelihoods by addressing issues such as climate change, food security and labor shortages through the use of digital technologies. The latter can increase economic benefits for farmers by reducing labor and resource costs, increasing yields and improving quality, with a 30.4%

increase in economic benefits for every increase in the intensity of digital technology use [3]. Technology promoters can use this information to increase farmers' awareness, highlight concrete benefits, and provide specialized agricultural services to stimulate the adoption of digital technologies applied to ground and aerial robots [4].

In response to these challenges, robots and autonomous machines are emerging as advanced solutions in precision agriculture. As robotics and autonomous machines are spreading into new agricultural contexts, technologies such as deep learning and machine learning are also finding increasing application, with increasingly sophisticated uses in different stages of agricultural production. In the case of complex and unstructured environments, such as curved paths and uneven terrain, depth vision sensors, such as stereo cameras, structured light cameras, and time-of-flight cameras, are the best solution to help robots accurately perceive the correct path of navigation [5,6]. A possible solution could be represented by human–robot collaboration approaches for agricultural tasks and applications [7].

Achieving high autonomous work performance is still the focus of agricultural robot research and development. In this regard, rapid hardware and software upgrading also provides important support for the advancement of technology in robot recognition and positioning methods [8,9], as well as in working more quickly and more gently than human labor when interacting with plants and animals [10].

These solutions allow for the realization of the main current applications of robotic systems in agriculture, such as pre-sowing soil preparation, planting, weeding and plant protection, harvesting, yield estimation and phenotyping. On the basis of general criteria, agricultural robots can be classified according to locomotion system, final application, presence of sensors, robotic arms and/or computer vision algorithms, stage of development and country and continent of origin, showing that most applications of robotic systems in agricultural environments are directed towards the development of 4WD robots, without a robotic arm, used for the removal of weeds, and they use RGB cameras [11,12]. Jin et al. [13] classified agricultural robots based on (i) type of industry (crop farming, livestock and poultry farming, aquaculture); (ii) function (phenotyping, monitoring, mapping, object handling, environment cleaning, health protection, etc.); (iii) intelligence level (remote-control, man–robot collaboration, full autonomous); (iv) working mode (selective, non-selective); (v) mobility (stationary, mobile); (vi) space (aerial, ground, aquaculture). Each classification shows that the industrial chain of agricultural robots is more complex than that of mechanical manufacturing, and several challenges still need to be addressed.

Against a wide range of solutions on the market, even today there is still a perceived gap between the current level of development of agricultural robots and the actual demand from farmers. The most impactful challenge to the implementation of automation and robotics on agricultural farms is the high cost of technology acquisition, which represents one of the bottlenecks in the sector [13]. In addition to the economic impacts of robots, the political, social, cultural and safety implications related to the introduction of robots should also be analysed, which have so far received little attention in the broader literature on agricultural robotics [14].

The regulation of autonomous robots remains a sensitive and pressing challenge that needs a community approach and the involvement of all actors in the field. The regulatory framework for the design and approval of agricultural machinery, ensuring safety for its intended use, consists of the Machinery Directive and the Mother Regulation—Directive 2006/42/EC (<https://eur-lex.europa.eu/eli/dir/2006/42/oj> accessed on 27 September 2024). These documents allow manufacturers to introduce machines with innovative technological solutions onto the market, even in the absence of reference regulatory standards, as long as they guarantee a level of safety equivalent to that of traditional machines. A robot

is defined as “an autonomous machine capable of completing a series of operations, such as driving, without the need for an operator on board”. In fact, the main characteristic of this category is independence in decision-making, allowing the robot to determine whether and when to stop in relation to an obstacle, or when and how to perform a specific operation in the field.

The first significant change will come with the entry into force on 20 January 2027 of the new Machinery Regulation EU 2023/1230 (<https://eur-lex.europa.eu/legal-content/IT/TXT/?uri=CELEX%3A32023R1230>), approved on 29 June 2023. This regulation will replace the previous Machinery Directive, introducing new safety requirements for autonomous agricultural machinery. One of the main changes is the introduction of the definition of autonomous machinery as “vehicles whose operational safety is ensured by autonomous functions”, i.e., systems “capable of self-evolution without permanent operator interaction”. With the new regulation, if a safety function is managed by a system capable of self-evolution, as is the case with most agricultural robots, it will no longer be sufficient to provide a self-declaration of conformity based on one’s own risk analysis. Instead, certification by an external body will be required. Furthermore, the concept of supervision and supervisor is introduced where supervision is defined as “non-permanent remote surveillance of the autonomous mobile machine”, to be carried out by the supervisor when the machine is operating in autonomous mode.

The rules of the new Regulation will be integrated and harmonised with future European standards on Artificial Intelligence and cybersecurity, to create a more comprehensive and up-to-date regulatory framework for today’s technological challenge that will only apply to robots possessing certain technologies. This could lead to certifications by third-party bodies, increasing development costs and making it more difficult for companies. At international level, the ISO 18497-1:2024 *Agricultural machinery and tractors—Safety of partially automated, semi-autonomous and autonomous machinery, Part 1: Machine design principles and vocabulary* (<https://www.iso.org/standard/82684.html> accessed on 21 October 2024) [15] has the scope “to assist in the provision of more specific safety requirements, means of verification and information for use to ensure an appropriate level of safety for agricultural machinery and tractors with partially automated, semi-autonomous and autonomous functions used in a specified way”.

But technological solutions are also needed to increase the capacity for control and accident prevention, developing increasingly advanced systems based on sensors, lidar cameras, RTK-GPS, and internet connection [16,17].

In order to meet required quality standard, some systems also allow farmers to monitor the quality of the work carried out by the robots in real time, having a history of the operations performed and intervening quickly in the event of anomalies [17]. An important aspect also concerns systems that allow robots to communicate with each other and with UAVs (Unmanned Aerial Vehicle) or devices in the field [2]. When multiple robots are used, even if one is down for repair and maintenance, others in the fleet can still continue to do the operation thereby minimizing the risk of halting the entire operation [10].

In 2019, a study was published in which, after conducting a search on various databases and narrowing the field from 4817 documents found to 18 because they were related to qualitative analyses, the economics of agricultural automation and mechatronic robotics were analysed and research gaps identified. Specifically, all of the studies found scenarios in which automation and robotics technologies were beneficial, but most of the studies employed partial accounting, taking into account only the costs and revenues directly affected by the introduction of automation or robotics and assuming that everything else remained unchanged. None analysed changes in farming systems or regional and national effects on markets, trade and labour demand. The review emphasised the need

for more research on the economic implications of these technologies. Moreover, all the studies analysed focused on agriculture in developed countries, while most of the world's most serious agricultural problems are found in developing countries, where the needs are therefore different [18].

So, agricultural robots face a number of challenges, both general and task-specific. General challenges include route planning, safety issues, in particular human detection, and the management of robot fleets. Activity-specific challenges are related to the unique requirements of crop structure, detection and classification of crops or pests, and the precise application of agricultural inputs. Many of these challenges stem from the search of optimal solutions in vision systems, in robotic actuation, in navigation in semi-structured environments, in the areas of battery technology, and in intelligence systems required to control both the robotic platform and its tools [19,20]. Another aspect of fundamental importance concerns the environmental sustainability of the use of agricultural robots [19]. As recalled in the from Farm to Fork strategy, it should be highlighted that robotics will also play a key role in decarbonizing agricultural production systems. The large-scale use of agricultural robots on farms can assume a key role to reduce compaction by lightweighting robotic machines [21], to optimise crop nitrogen use and reduce N₂O emission [22,23], to reduce farm waste, including losses from pests and disease by means of Artificial intelligence and machine learning [24], to reduce carbon emissions by the use of electrified robotic vehicles [25,26]. To achieve these goals, it becomes essential to place attention on the integration of technical systems with social and ecological ones during the design, production and use of robotic technology [19].

In this context, the paper reviews the main robotic solutions already on the market or still at the prototype/design stage that have been able to overcome all these challenges and that can be used in the field. This work stems from the need to learn about technological advances in agricultural robotics and is part of the Research Project of Significant National Interest (PRIN2022) PNRR “ROVERCROP—Sustainable Implementation of Unmanned Ground Vehicles for On-field Agricultural Activities” (prot. n. P2022R8YT9_002), coordinated by the University of Sassari.

2. Materials and Methods

Many companies have invested and continue to invest in the development of increasingly precise and high-performance robotic solutions that meet the needs of the agricultural sector [27–135] as shown in Table 1. Precisely, the data reported in this paper are not derived from literature research, but were extrapolated directly from data sheets available online of robots that can already be purchased or of prototypes that can be tested directly in the field by farms that request them. For these reasons, some information regarding technical characteristics of the robots are not directly accessible online.

Because companies use various websites and trade magazines to advertise their products and present the results of their research and development work, the survey presented in this article was conducted through Agtecher's website, analyzing trade magazines and consulting agricultural trade show websites. Participating in agricultural fairs and exhibitions and organizing field tests of their robots is another means used by manufacturers to directly reach the end consumer. For this reason, with the aim of including most of the robots already produced to support agriculture, reports from major trade shows dedicated to agricultural automation were researched and analyzed. By adopting this survey method, it was possible to obtain an overview of the agricultural robots that manufacturers/companies have made showing some technical choices and their main characteristics in a single table (Table 1).

Table 1. Overview of online robot.

Company	Robot Name	State	Function	Crops	Engine	Locomotion	Weight (kg)	Dimensions (cm)	3-Point Hitch	PTO	Thermal Autonomy	Battery Autonomy	Power (kW)
Agreenculture [27]	Ceol	FR	Tool-carrier	Open field crops	Hybrid Diesel engine Electric	Tracks		170 × 72 or 120 × 21	yes		>20 h	60–90 min	10
ACFR [28]	SwagBot	AU	Pasture	Open field crops	Electric (PV)	Tyres			no				
	Rippa		Chemical weeding	Vegetable crops	Electric (PV)	Tyres							
	Ladybird		Monitoring	Vegetable crops	Electric (PV)	Tyres							
	Mantis		Monitoring	Ochards	Electric (PV)	Tyres							
	Shrimp		Monitoring	Ochards	Electric (PV)	Tyres							
Adigo [29]	Kilter AX-1	NO	Chemical weeding, Plant protection	Vegetable crops	Hybrid Diesel engine- Electric	Tyres							
Advanced Farm Technologies, Inc [30]	–	US	Harvesting	Vegetable crops	Electric	Tyres							
Afara Agricultural Robot [31]	Afara-cotton	TR	Harvesting (cotton)	Open field crops	Electric	Tyres							
Agerris	Digital Farmhand Platform	AU	Mechanical weeding	Vegetable crops	Electric (PV)	Tyres							
Agricobots [32]	ELETTRA ST-2045	IT	Weeding	Ochards	Hybrid Diesel engine- Electric	Tracks		190 × 115 × 120	no				
Agrobot [33]	E-series	ES	Harvesting	Open field crops	Electric	Tyres							
	Bug Vacuum		Plant protection	Vegetable crops	Diesel engine	Tyres	1270		no		15 h	-	153
Agrointelli [34]	Robotti	DK	Tool-carrier	Vegetable crops	Diesel engine	Tyres	2850	244 × 305 or 490 × 215	yes	optional	60 h	-	54
AgXeed [35]	AgBot	NL	Tool-carrier	Open field crops, Ochards	Hybrid Diesel engine- Electric	Tracks				optional			
AI.Land [36]	Etarob	DE	Ploughing, Seeding,	Vegetable crops	Electric (PV)	Tyres							

			Irrigation, Fertilization, Harvesting										
Aigro [37]	Aigro UP	NL	Weeding, Mowing	Ochards	Electric	Tyres		130 × 60 × 60		-	10 h		
Amazona [38]	Bonirob	DE	Plant protection, Monitoring	Open field crops, Vegetable crops	-	Tyres							
Amos Power [39]	Amos Power A3/A4	US	Tool-carrier	Open field crops	Electric	Tracks	3040	295 × 180 × 160	yes	yes	-	8 h	55–62
Andela-TNI [40]	Andela ARW- 912	NL	Weeding	Vegetable crops	Electric (solar e batteries)	Tyres							
- [41]	MoonDino	CH	Weeding	Open field crops	Electric (PV)	Tyres							
Augean Robotics Inc. [42]	Burro	US	Transport	Open field crops, Ochards	-	Tyres							
AutoAgri [43]	ICS 20	NO	Tool-carrier	Open field crops	Electric or Hybrid Diesel engine- Electric	Tyres			yes	yes			
Autonomous Tractor Corporation [44]	Spirit (demo)	US	Plant protection	Open field crops	-	Tracks	11,340						
Autopickr [45]	Gus	UK	Harvesting (asparagus)	Vegetable crops	Electric	Tyres							
AutoSaturn	-	AR	Soil tillage	Open field crops	Electric	-							
AvL Motion bv [46]	Compact S9000	NL	Harvesting (asparagus)	Vegetable crops	-	Tyres	4500	600 × 236 × 293					
Avrora Robotics [47]	AgroBot	RU	Tool-carrier	Open field crops, Ochards	-	Tyres							
Berg Hortimotive [48]	Meto	NL	Plant protection	Vegetable crops	Electric	Tyres							
	Plantalyzer		Monitoring	Vegetable crops	Electric	Tyres							
Black Shire S.R.L. [49]	Robot RC 3075	IT	Tool-carrier	Ochards	Diesel engine	Tracks			yes				
Bobcat [50]	RogueX	US	Seeding, Weeding e Monitoring	Open field crops	Electric	Tracks							

Cambridge Consultants [51]	Mamut	UK	Monitoring	Open field crops, Orchards	-	Tyres							
Carré SAS [52]	Anatis	FR	Mechanical weeding	Open field crops	Electric	Tyres			yes		-	8 h	
Case IH [53]	Magnum	US	Autonomous elettric tractor	Open field crops, Orchards	-	Tyres							
Cerescon BV	Sparter	NL	Harvesting (asparagus)	Vegetable crops	-	Tyres							
CET Electronics [54]	Rovitis 4.0	IT	Plant protection	Ochards	Diesel engine	Tracks							
Clearpath Robotics [55]	Husky	FR	Monitoring	Open field crops, Orchards	Electric	Tyres	50	99 × 67 × 39			-	3 h	
	Warthog		Monitoring	Open field crops, Orchards	Electric	Tyres	280	152 × 138 × 83			-	3 h	
Continental AG [56]	Contadino	DE	Monitoring	Open field crops, Orchards	Electric	Tyres							
Dawn Equipment [57]	-	US	Tool-carrier	Open field crops, Orchards	Electric	-							
Deepfield Robotics	Aquila	DE	Mechanical weeding	Open field crops	Electric	Tyres							
Denso [58]	Faro	JP	Harvesting	Vegetable crops	Electric	Tyres							
DFKI [59]	Shiva	DE	Harvesting (strawberry)	Open field crops	Electric	Tyres	150	245 × 120 × 100					
Digital Workbench [60]	Tipard 350	DE	Transport	Open field crops	Electric or Hybrid Diesel engine-Electric	Tyres	350	220 × 120 or 170 × 110–130					
	Tipard 1800		Tool-carrier	Open fieldcrops	Electric or Hybrid Diesel engine-Electric	Tyres	2600	425 × 175 × 185			24 h (hybrid)	12 h	
Direct Machines [61]	Land Care Robot	US	Tool-carrier	Ochards	Electric (PV)	Tyres	635	203 × 127	yes	yes	-		45
Dogtooth Technologies [62]	–	US	Harvesting (strawberries)	Vegetable crops	Electric	Tracks	400	180 × 70 × 200	-	-	-	22 h	

DOT Farming reimagined [63]	OmniPower	CA	Weeding, Seeding, Harvesting	Open field crops	Diesel engine	Tyres							
Earth Automation [64]	Dood	IT	Tool-carrier	Open field crops, Orchards	Diesel engine	Tracks		yes	yes	10 h	-	59	
EarthSense Inc [65]	TerraSentia	US	Monitoring	Open field crops, Orchards	-	Tyres					3 h		
ecoRobotix SA [66]	AVO	CH	Chemical weeding	Open field crops	Electric (PV)	Tyres							
Ekobot [67]	Gen-II	SE	Mechanical weeding	Open field crops	-	Tyres							
Elatec [68]	E-tract	FR	Mechanical weeding	Open field crops	Electric	Tyres	<800	300 × 170	yes		-	4–8 h	
Exel Industries [69]	Exxact Robotics Traxx	FR	Tool-carrier	Ochards	Electric	Tyres							
Exobotic Technologies [70]	Exobot Land-A2	BE	Tool-carrier	Ochards	Electric	Tyres	250				4 h		
FarmBot [71]	-	US	Urban farming	Open field crops, Vegetable crops	Electric	-							
FarmDroid ApS [72]	FD20	DK	Seeding, Mechanical weeding	Open field crops, Vegetable crops	Electric (PV)	Tyres	900				18–24 h		
Farmertronics Engineering bv [73]	eTrac-20	NL	Autonomous electric tractor	Open field crops, Orchards	Electric + Hydrogen	Tyres						35	
FarmWise [74]	–	US	Weeding	Open field crops	-	Tyres							
Fendt [75]	Mars	DE	Seeding	Open field crops	Electric	Tyres	50						
	Xaver		Seeding	Open field crops	Electric	Tyres	150–250						
FFRobotics [76]	FFRobot	IL	Harvesting (apples)	Ochards	-	Tyres							
FieldRobotics SRL [77]	HammerHead FR-01	IT	Tool-carrier	Ochards	Electric	Tracks	600		yes	yes	-	8 h	10
FieldWork Robotics [78]	Fieldworker 1	UK	Harvesting (raspberries)	Vegetable crops	-	Tyres							
Fox Robotics [79]	Hugo RT Gen. III	US	Transport	Open field crops, Orchards	Electric	Tyres		107 × 63					

Franklin Robotics	Tertill	US	Mechanical weeding	Open field crops, Orchards	Electric	-							
Free Green Nature [80]	Icaro X4	IT	Plant protection	Ochards	Diesel engine	Tyres							
Grape [81]	–	ES	Monitoring	Ochards	Electric	Tyres							
GreenPatrol	GreenPatrol robot	NL	Monitoring	Vegetable crops	Electric	Tyres							
GUSS Automation LLC [82]	GUSS	US	Plant protection	Ochards	-	Tyres							
H2arvester System BV [83]	H2arvester	NL	Harvesting	Open field crops	Electric (PV)	Tyres							
H2L Robotics [84]	Selector 180	NL	Plant protection	Open field crops	-	Tracks							
Hari Tech [85]	HariBOT	HU	Tool-carrier	-	-	Tracks							
Harvest Automation [86]	HV-100	US	Transport	Vegetable crops	Electric	Tyres	45	61 × 53		-	4–6 h		
Harvest Croo Robotics [87]	Harvester B5.1	US	Harvesting (strawberries)	Vegetable crops	-	Tyres							
Hortibot	–	US	Mechanical weeding	Vegetable crops	-	Tyres							
Ibex Automation Ltd. [88]	–	UK	Chemical weeding	Open field crops, Orchards	Electric	Tracks							
IdaBot	–	US	Plant protection	Ochards	-	Tracks							
Improvis [89]	Citrus Harvesting Robot	AM	Harvesting	Ochards	-	Tyres							
Inesc Tec [90]	Modular-E	PT	Tool-carrier	Ochards	Electric	Tyres							
	Weta Robot		Monitoring, Weeding, Harvesting	Ochards	Electric	Tyres							
InsightTRAC, LLC [91]	InsightTRAC	US	Plant protection	Ochards	Electric	Tracks							
Instar Robotics [92]	Trooper	FR	Transport	Nursery plant	Electric	Tyres	65	70 × 60 × 65	-	-	-	10 h	-
Iron Ox	Grover	US	Transport	Vegetable crops	Electric	-							

Jacto [93]	Arbus 4000 JAV	BR	Plant protection	Ochards	Diesel engine	Tyres							97
Kilter System [94]	AX-1	NO	Weeding	Vegetable crops	Electric	Tyres	260						
Korechi [95]	RoamIO-HCW	CA	Seeding, Weeding, Trasport, Mowing, Monitoring	Ochards	Electric	Tyres	590	187 × 179 or −212 × 162					10
	RoamIO-HCT		Seeding, Weeding, Trasport, Mowing, Monitoring	Open field crops	Electric	Tracks	450	160 × 170 × 127					10
Krone e Lemken [96]	Combined Powers	DE	Tool-carrier	Open field crops	Hybrid Diesel engine-Electric	Tyres			yes	yes			170
Kubota [97]	New Agri Concept	JP	Autonomous elettric tractor	Open field crops, Ochards	Electric	Tyres			yes				
	KFAST		Weeding	Ochards	Electric	Tyres							
Lambers ed Exobotic Technologies	WTD4	NL	Weeding	Open field crops	Electric	Tyres	1110	300 × 150	yes	yes	-	4+ h	20–40
Lj Tech [98]	Orchard S450	CN	Weeding	Ochards	Hybrid Gasoline engine-Electric	Tracks		210 × 120 × 120					
Maka Autonomous Robots	Maka-ARS	US	Mechanical weeding	Open field crops, Vegetable crops	Diesel engine	-							
Merlo [99]	Cingo M600A-e	IT	Tool-carrier, Weeding	Ochards	Electric	Tracks							
Meropy [100]	SentiV	FR	Monitoring	Open field crops	-	-	15	-					
Metalfor [101]	VAX	AR	Weeding, Seeding, Fertilization, Harvesting	Open field crops	Diesel engine	Tyres							113

Metazet Formflex [102]	IRIS	NL	Transport, Plant protection	Vegetable crops	-	-					
Metomotion [103]	GRoW	IL	Harvesting, Monitoring	Vegetable crops	-	-					
Mineral (Alphabet)	Mineral Rover	US	Monitoring	Vegetable crops	Electric (PV)	Tyres					
Mobile Autonomous Systems and Cognitive Robotics [104]	Etarob	DE	Mechanical weeding	Open field crops	Electric	Tyres					
Naio Technologies [105]	Oz	FR	Tool-carrier	Open field crops, Vegetable crops	Electric	Tyres	150	130 × 47 × 83		8 h	
	TED		Weeding	Ochards	Electric	Tyres	1905	400 × 190 × 240 or – 275		8 h	
	Orio		Weeding	Open field crops, Vegetable crops	Electric	Tyres	1450	-			
	Jo		Tool-carrier	Ochards	Electric	Tracks	850			8 h	
Nexus Robotics [106]	La Chèvre	CA	Weeding	Open field crops	Hybrid Diesel engine-Electric	Tyres					
Octinion Technology Group [107]	Rubion	BE	Harvesting (strawberries)	Open field crops	-	Tyres					
	Titanion		Tool-carrier	Open field crops	-	Tyres					
	Lumion		Weeding (strawberries)	Open field crops	-	Tyres					
	Fluxion		Transport (strawberries)	Open field crops	-	Tyres					
Odd.bot [108]	Maverick	NL	Mechanical weeding	Open field crops	Electric	Tyres					
Osiris Agriculture [109]	Oscar	FR	Irrigation, Fertilization	Open field crops	Electric	Tyres					
Oxin [110]	Smart Machine	UK	Tool-carrier	Ochards	Electric	Tracks					99
PeK automotive [111]	Agilehelper	SI	Tool-carrier	Ochards	Electric	Tracks					
	SlopeHelper		Tool-carrier	Ochards	Electric	Tracks					

Pixelfarming Robotics [112]	Robot One	NL	Mechanical weeding	Open field crops	Electric	Tyres	2140	240 × 374 or –534 × 223					
Polariks [113]	VitoScanner	FR	Monitoring	Ochards	-	Tyres							
Precision Makers [114]	GreenBot	NL	Tool-carrier	Open field crops, Ochards	-	Tyres			yes				
Raussendorf GmbH [115]	Casar	DE	Tool-carrier	Open field crops, Ochards	-	Tyres	1500	300 × 130 × 92	yes	yes			
Renu Robotics Corp. [116]	–	US	Tool-carrier	Open field crops, Ochards	Electric	Tyres							
Robotics Plus [117]	Prospr	NZ	Tool-carrier	Open field crops, Ochards	Hybrid Diesel engine-Electric	Tyres							
Robotnik [118]	Vinbot	ES	Monitoring	Ochards	-	Tyres							
Sabi Agri [119]	Zilus	FR	Tool-carrier	Ochards	Electric	Tracks					10 h		
Saga Robotics [120]	Thorvald	NO	Plant protection	Open field crops, Ochards	Electric	Tyres							
Sitia [121]	Trektor	FR	Tool-carrier	Ochards	Hybrid Diesel engine-Electric	Tyres	3200	variable	yes			24 h	8
SIZA Robotics [122]	Toogo	FR	Tool-carrier	Vegetable crops	Electric	Tyres			yes				
Small Robot Company (SRC) [123]	Tom	UK	Weeding, Monitoring	Open field crops	-	Tyres							
SoftiRover	e-K18	FR	Tool-carrier	Open field crops	Electric	Tyres							
Solinftec [124]	Solix	US	Weeding	Open field crops, Vegetable crops	Electric	Tyres							
SwarmFarm Robotics [125]	Swarmbot Juliet	AU	Weeding	Ochards	Diesel engine	Tyres							
	Indigo		Tool-carrier	Open field crops	Electric	Tyres							
TartanSense [126]	BrijBot	IN	Weeding	Open field crops, Vegetable crops	-	Tyres							
Tevel Agricultural Technologies [127]	Flying Harvest Robots	IL	Harvesting	Ochards	-	-							
Tortuga AgTech [128]	–	US	-	Vegetable crops	-	Tyres							

Trabotyx [129]	Trabotyx	NL	Mechanical weeding	Open field crops, Vegetable crops	Electric (PV)	Tyres				4 h		
Traptic	–	US	Harvesting (strawberries)	Vegetable crops	-	Tyres						
TRIC Robotics [130]	Eden	US	Weeding	Vegetable crops	Electric or Hybrid	Tyres						
Vinescout [131]	–	FR	Monitoring	Ochards	Electric (PV)	Tyres						
VitiBot [132]	Bakus	FR	Weeding, Soil tillage	Open field crops, Ochards	Electric	Tyres	2050	350 × 175 × 200		-	10 h	
Vitirover [133]	Vitirover	FR	Mechanical weeding	Open field crops, Ochards	Electric (PV)	Tyres	18	74 × 38 × 28				16
Yanmar Agribusiness Co. Ltd. [134]	YV01	JP	Weeding (vineyard)	Ochards	Gasoline engine	Tracks						18
Zauberzeug [135]	Field Friend	DE	Monitoring	Open field crops	Electric (PV)	Tracks						

The table that includes all the information collected consists of 14 columns and has been sorted alphabetically by the name of the company that manufactures the robot so that it can be immediately identified through the website reference. In the second column is the name of the robot, in the third the country in which the company is located, and in the following columns the function of the robot, the crop on which it can be used, the type of engine, and the means of locomotion used. This information is available for all robots found through the different channels investigated. As for the data reported in the other columns regarding robot weight, size, presence of three-point hitch and PTO, motor or battery autonomy, and motor power, it was not possible to find the complete information for all the robots listed in Table 1. However, it was chosen to report them in the table. In particular, technical information could not be found for all the robots discovered online and examined because manufacturers do not make all data on their product accessible, but do give the option of requesting a demo eventually. For example, it was also not possible to access the selling prices of all robots because they were available for only a very few examples in the sites examined and therefore were not included in the table.

It should be emphasised that all the examples given in the Table 1 are autonomous agricultural robots, intended exclusively for the open field or greenhouses: the numerous examples of robots used in livestock farms and all machines equipped with an operator's driving position as well as robots for aquaculture, food processing and packaging have been excluded.

The next step was the graphic processing of all data, resulting in 5 diagrams to optimise access to information and to comment them by topic.

First, the data were grouped according to the country of origin of the producing company to get an idea of the worldwide spread of agricultural robots. The analysis of robotic solutions for agriculture was conducted by grouping data from the perspective of functions performed, robots' intended use, type of power supply, locomotion system, and performance characteristics (weight, size, three-point hitch, PTO, autonomy, and power). The analysis conducted was useful to understand whether the solutions on the market were sufficiently varied or whether all companies were following the same trend. In addition, special attention was paid to the type of power supply as it was crucial to assess whether manufacturers were sensitive to the environmental sustainability aspects of these technological solutions.

The research is updated to the year 2023–2024 and includes the robots proposed so far; however, it should be noted that with the accelerated pace of new robot development in recent years, it cannot be ruled out that some may be missed.

3. Results and Discussion

In order to obtain an up-to-date and comprehensive overview of the agricultural robots available to date, research has been quite challenging, mainly because sources are heterogeneous, information regarding technical characteristics is either missing or not well defined on the sites where they are offered. In addition, it was not easy to distinguish robots already on the market from those still in the design and/or prototype stage.

With reference to the Table 1, 145 robots created by 123 companies were identified [27–135]. For some robots, the company's website no longer exists, but they were still considered.

As mentioned above, only information on geographical distribution of the companies, function performed by agricultural robots, reference crop, engine type, and locomotive system is always available online. Weight, dimensions, presence of 3-point hitch and PTO, battery autonomy, and power are available for only a few cases (Table 1).

Starting from geographic distribution by State (Figure 1), the manufacturers that have shown the most interest in new agricultural robotic technologies are concentrated in

Europe (about 63%) and America (about 28%) while Asia (about 8%) and Oceania (about 1%) represent only a low percentage. In any case, this article shows the state of the art updated to the last two years, at present, there is still a great distance between the current development level of agricultural robots and the actual demand for agricultural production [13].

In Europe, namely France (about 14%) and Netherlands (about 13%) are closely followed by other European countries, including Germany (about 9%), Italy (about 6%) and Great Britain (about 5%) while other countries remain under the 4%. The United States together with Canada accounts for more than 30% while South America reaches about 3%. Among Asian countries ranked first are Japan (about 3%) and Israel (about 3%).

The data confirm that the higher level of agricultural innovation coincides with greater interest in the implementation and dissemination of agricultural robots. In many countries, the push for increasing production of robotic solutions in agriculture is driven on the one hand by the quest for greater productivity and at the same time by an attempt to cope with the growing shortage of labor, especially seasonal labor. Labor is the main limiting factor for agricultural development in both developed and developing countries. Labor shortages drive the search for new robotic solutions to secure revenues and maintain high productivity. The high manual labour requirement increases the demand for robotics and automation. Conversely, in countries where labor wages are low and the environments are poorly structured, the use of robots is less attractive for farmers and manufacturers [20]. At the same time, robotics and automation require skilled labor and more expensive equipment, even if they can contribute to increasing agricultural productivity while the labor cost decreases sufficiently to counterbalance the higher initial cost of robots [26].

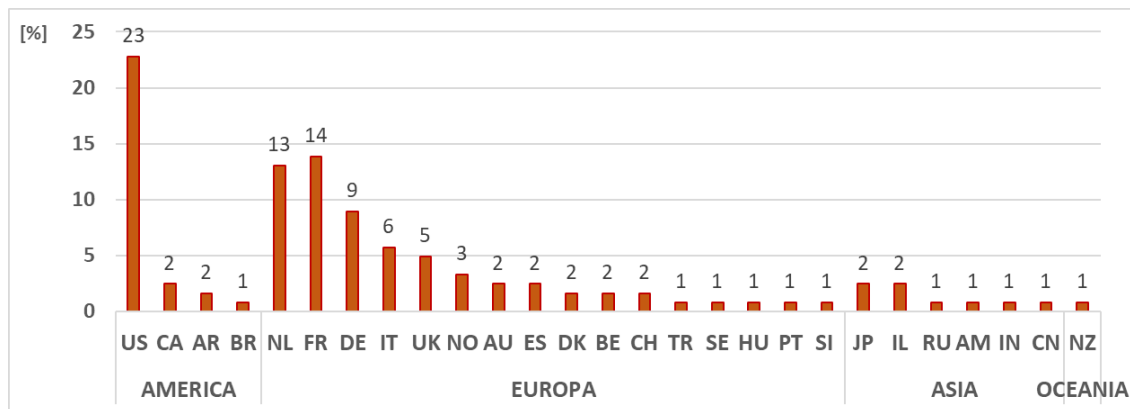


Figure 1. Geographic distribution by State (values in percentage).

The survey denotes that there are already robotic solutions capable of performing all different field operations from weeding to soil tillage with a little percentage of undefined function (Figure 2). In particular, manufacturing companies have directed their interest toward robots for weed control (26%), especially chemical and mechanical weeding. Weeding and plant protection represent the most onerous processing for workers because it involves handling and exposure to substances that are harmful to humans especially in confined spaces [136]. To better perform this operation, sensors and equipment are mounted on these robots to optimize the use of inputs. In the future, autonomous robotic weed control systems may provide a cost effective alternative for herbicides and labour by integrating both GPS based navigation and machine vision-based relative navigation [137–139]. Some robots are able to performs the tasks of detecting (through cameras and ultrasonic sensors) and removing weeds, not only using mechanical tools, but also spraying herbicide [140].

A rather large number of robots are used as tool carriers (19%), ideal for work in areas with extreme strong slope conditions dangerous to the operator. Equally numerous (14%) are the robots dedicated to the monitoring crops or soil parameters. One of the activities in which technology was first employed is definitely monitoring. Since artificial intelligence and machine learning use, algorithms capable of studying insects, diseases, and plant vigor have been developed to help farmers in farm management because, combined with automation, they make interventions more timely. Although robotic monitoring technology is developing so rapidly, and so many types of sensors are integrated to get big data, accurate and efficient autonomous monitoring remains a challenge. The next step will be to further improve the friendliness of big data to farmers and to give direct decisions to end-users. Between the different crops, autonomous vineyard monitoring robot can help wine producers to measure fundamental vineyard parameters and to control pests disease [13]. Other examples included a robotic platform for data acquisition and yield estimation in orange and sugarcane crops by using specific and highly reliable sensors [11].

The harvest is also well represented especially for cotton, raspberries, strawberries, apple and vegetable. One of the issues affecting the agricultural sector today is the lack of manpower, employed especially during the harvesting of fruits and vegetables which requires a lot of hours of agricultural works. Experimenting with platforms that can replace farm operators requires more effort, as harvesting methodologies must be devised that can assess the degree of ripeness, not damage produce during handling, and work while subjected to numerous environmental variables. The main challenges yet to be addressed in the development of these systems involve guiding the robot through the field from tree to tree and row to row, locating fruit, catching and releasing selected fruits or vegetables [136,141]. Although robots have not yet achieved sufficient results to completely replace the human picker in terms of harvest success rate, some challenges need to be addressed to solve the problem of fruit occlusion and changes in environmental lighting also by improving the computer vision algorithms used [11].

A smaller percentage of robots were made to plant protection treatments (8%), seeding (5%) and transport (6%). Although chemical risks to the operator are reduced with robots, there are only a few examples capable of performing crop defense, and some of these are still being tested. A very small percentage of robots (1–2%) are capable of performing other crop operations such as fertilisation, mowing, soil tillage and irrigation. Their use is still economically and managerially unprofitable for these functions. For in-row spraying, tilling, ditching, etc., in fields, greenhouse, the navigation is still a research perspective [142,143]. The automation of crop operations such as sowing, irrigation, fertilisation and tillage have attracted less interest because precision agriculture, specifically automated and semi-automated guidance and variable-rate operating machines, has virtually automated these operations, and in fact the farmer who has invested in 4.0 technologies will be less inclined to further farm conversion. In addition, in the case of seeding, the presence of the seed and therefore the adoption of technical solutions that increase the weight and bulk of the structure must be anticipated. Moreover, the size of the planting tray, the vacuum chamber pressure and the planting speed were the main factors that directly can affect the seeding quality [144,145]. Conversely, the use of robots could decrease pressure on the soil by reducing the phenomenon of compaction along with the other general benefits already discussed for other crop tasks [11].

Regarding fertilisation, there is to be considered an increased demand for power from the robot, which can be directed to the power take-off, if arranged, for fertiliser release. Some solutions provide separate reservoirs which allow for an exclusive tank for herbicides and/or fertilizers. Today, these robots are still complex, slow, and costly to be made available to farmers even if they could be useful for cost saving and human labour safe

[146]. Few robots are dedicated to new functions such as urban farming (>1%) and pasture monitoring (>1%) probably because of a low cost/benefit ratio.

In general, high interest has been shown in robotic platforms capable of performing multiple types of functions because a single machine able to implement multiple processing can better justify the investment of financial resources in the purchase of a robot by the farmer. Moreover, robots have been employed in confined environments such as barns or greenhouses for several years, so now the next challenge is to make robots available that can also operate in more heterogeneous environments.

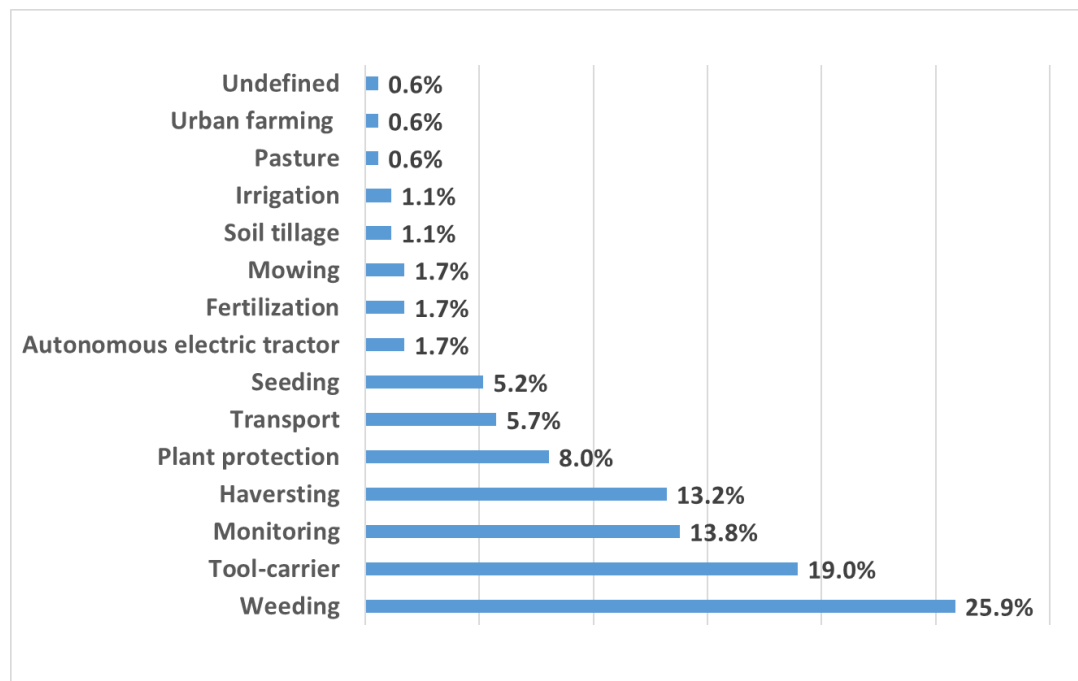


Figure 2. Functions performed by agricultural robots.

The types of crops that can be managed through the use of robotics are numerous. Although for most of the examples given in Table 1, the type of crop the robot is intended for is not indicated but only the context (arable, horticultural, tree), some of them are specific to certain crops, such as the company's Afara-cotton Agricultural Robot for cotton harvesting [31], others for asparagus harvesting [45,46], and still others for strawberry harvesting, weeding and transport [59,62,87,107].

Most of the robots are for herbaceous crops then for managing arable crops because the lack of obstacles makes it easier and safer to automate crop operations (Figure 3).

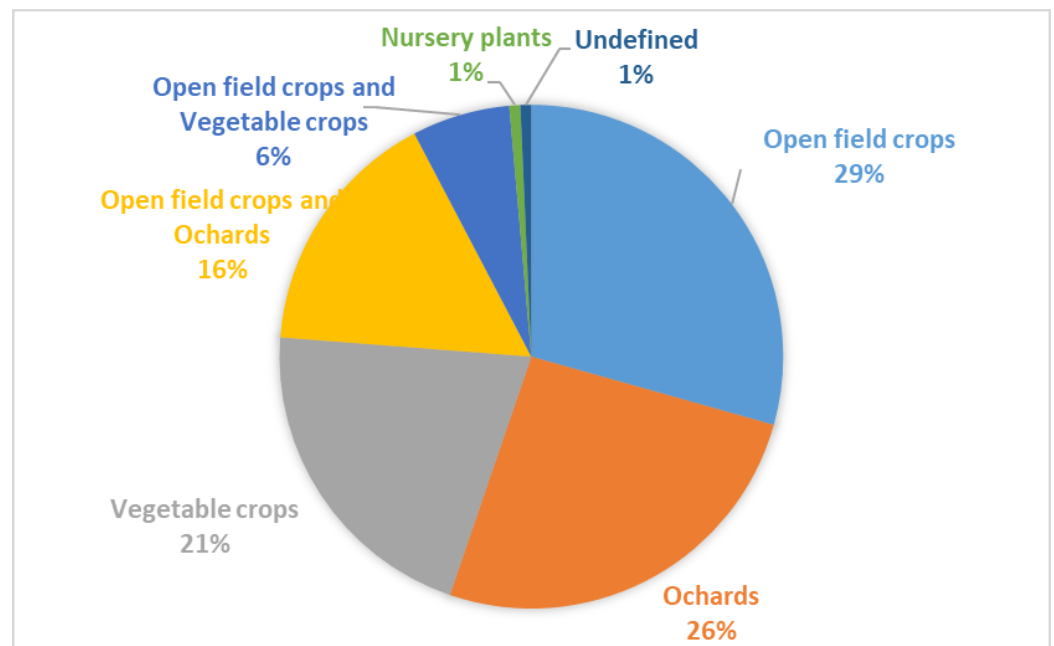


Figure 3. Crops managed by agricultural robots.

However, taking into consideration the examples given in the Table 1, the numerical difference with robots targeting orchards (26%) and open field crops including greenhouse crops (29%) is negligible, as shown in Figure 3. The possibility of use for vegetable crops is also quite widespread (21%). Due to technological innovation it is possible to equip robots with sensors that recognize rows, supports and any obstacles present and correct the trajectory.

Among tree crops, the vineyard is the one where automation is most present, both because of the regular structure of the planting and because it is among the most profitable productions, so as to incentivize the purchase of a robot [81,131–134].

Also noteworthy is the versatility of some robots, since the technical and structural characteristics allow them to be used in different environments, such as open field crop and orchards or open field crops and vegetable crops, more justifying the investment by farmers whose farm has multiple production addresses.

The analysis conducted shows that environmental sustainability, now a fundamental requirement in all production processes, is also pursued by the designers of manufacturing companies: more than half of the robots in Table 1 are electric, powered by accumulators (43%) or photovoltaic panels (PV) (12%). The graph in Figure 4 undoubtedly reflects today's situation regarding the ecological transition, with the abandonment of non-renewable energy sources, derived from petroleum, to take advantage of other energy sources, such as solar. Only 8% of the agricultural robots shown in Table 1 are powered by diesel engines, other 6% are powered by hybrid diesel-electric engine and about 1% by hybrid gasoline-electric or gasoline engine.

Agricultural robots, with their electric or hybrid motorization, can contribute to sustainable agricultural practices by reducing energy consumption from fossil fuels and GHG emissions. These solutions also translate into an advantage in terms of performance. In fact, it should be noted that while the efficiency of an endothermic motor is rather low, it varies between 80 and 90 percent in electric motors. In addition, an electric motor does not need to warm up to reach its maximum speeds, which makes it easier to increase or decrease speed. The mechanics of electric vehicles are simpler and with fewer components than endothermic, resulting in more compact and lighter vehicles with more power per unit weight.

Economically, the price per kilogram of lithium batteries has been steadily decreasing while the cost of fuel in recent years has also increased due to contingent factors. It should also be pointed out that the price of lithium batteries has been steadily decreasing unlike that of diesel fuel, which has shown an upward trend in recent years and whose value depends on multiple factors.

In the face of these advantages, reducing the size and weight of batteries and increasing their autonomy pose additional challenges for improving performances. In the case of electric robots, the first concern on the part of a buyer is surely the working autonomy or the power used to carry out determined operations. These can be partially overcome with the use of hybrid machines, in which the advantages of both diesel or gasoline and electric can come together. While, in rarer cases, the company offers the option of having the same robot with an exclusively electric or thermal motor.

One innovative solution is a robot equipped with a high-power fuel cell and high-power batteries, capable of delivering up to 35 kW. It is also equipped with two tanks of just over 9 kg of hydrogen, enough to provide an uninterrupted working range of up to 12 h. By using hydrogen as fuel, this solution not only provides a zero-emission alternative but is also a lighter, quieter machine that works efficiently and can be refueled quickly [73].

In relation to performance, the machines enjoy different autonomies (from 60–80 min to 24 h), mainly attributable to the different types of crop practice due to the different energy requirements. Some companies have tried to overcome the problem of low autonomy with rapid recharging speed or additional battery packs to minimize downtime.

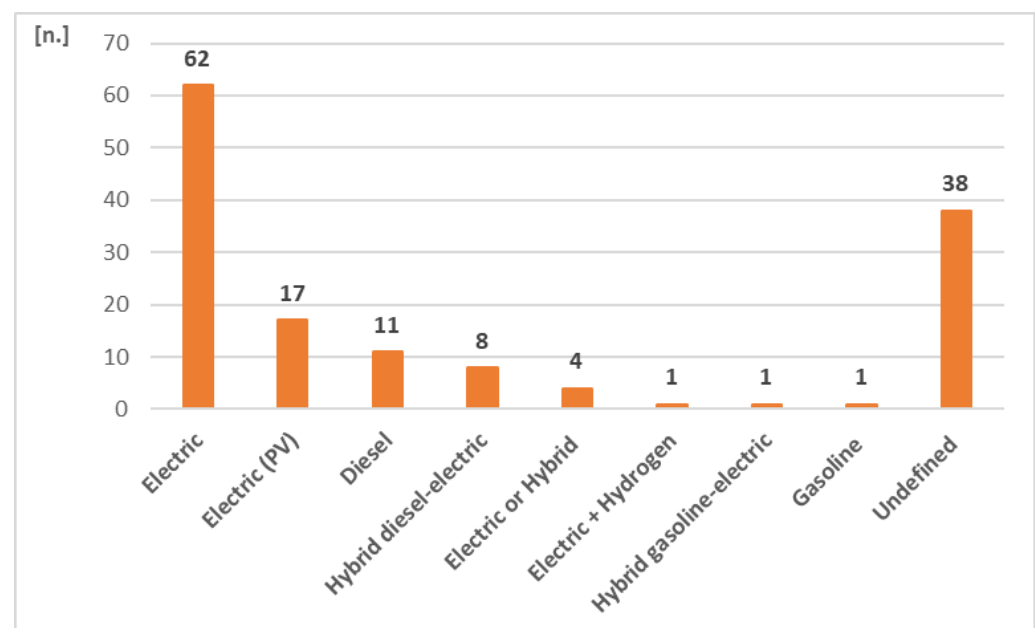


Figure 4. Power supply.

As for locomotion system, the choice of tires (75%), especially 4WD, prevails over tracks (18%), either iron or rubber (Figure 5). Although tracks boast greater stability and less soil compaction [21], the reasons why companies have decided to mount tires in their robots can be essentially two: the cost and the bulk [11]. A crawler track is made up of more parts, and the space it takes up is certainly larger than a tire. In addition, the track can be placed below the machine frame, so it must be taken into account that the adoption of tracks involves a widening of the track, which can be a disadvantage if we refer to robots for orchards or greenhouse crops.

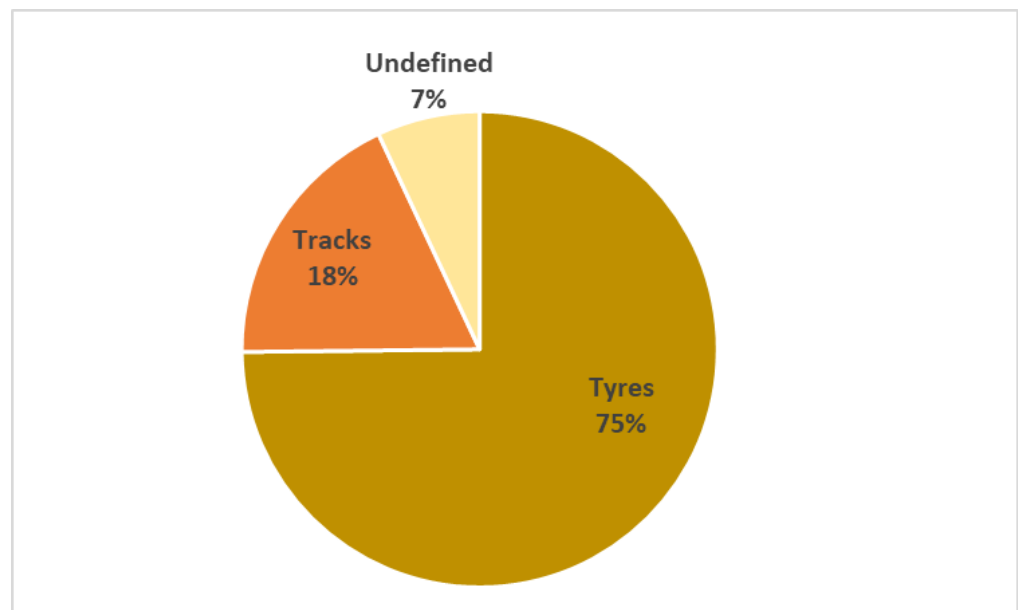


Figure 5. System of locomotion.

From the point of view of structural and performance characteristics, the lack of data did not allow the different robots to be studied in detail, as shown in Table 1. In general, it can be stated that robots have varied sizes and weights, so they can respond well to the needs of different agricultural landscapes although very often in heterogeneous environments they do not always succeed in adapting. The weight of a robot can range from a minimum of 15 kg to a maximum of 11,340 kg, with an average weight of 1405 kg. The dimensions, where indicated, are also very different from each other as is the rated power of the motor. The latter could range from a minimum of 8 kW to a maximum of 170 kW. Finally, not all robots are equipped with three-point attachments or power plugs. Robots that specialize in one or two crop operations fall into this group, while robots that have the provision to carry or operate an operating machine make up the group of more general machines (such as autonomous tractors or implement-carrying machines).

4. Conclusions

Field operations in agriculture are highly complex, and several challenges must be addressed to ensure a smooth transition into the era of robotics. The development of a robotic solution requires a comprehensive analysis of field operations and environmental context, along with a cost-benefit assessment [122,123].

On the other hand, one of the challenges is the difficulty of adapting to heterogeneous contexts within the same farm or among consortium farms.

At the same time, regulatory uncertainty regarding the use of self-driving machines in the field also remains. Robots are a resource for making agriculture more sustainable from an economic, social and environmental point of view, but to use them safely and effectively, appropriate regulation and advanced technology are needed to ensure reliable interaction with operators. One of the major criticisms of the regulatory framework in the making is that the standards are across-the-board, generalist, and therefore do not provide manufacturers with specific design requirements for Artificial Intelligent controlled safety devices. The new regulations will have a great impact on the design of agricultural robots, but we are still a long way from having production standards.

The lack of clear legislature in this regard also worries stakeholders in the sector. Until the relevant bodies delineate guidelines about the use and responsibilities of these new autonomous machines, their use in the field will always be limited. Wanting to give

a practical example in relation to operator welfare, if the operator is forced to control the robot during field operations, as is done with drones, it means that the operator has to follow it on foot between the rows. Certainly this is not advantageous, due to exposure to chemicals, in the case of operations such as weeding, or due to high or low environmental temperatures. Thanks to tractors equipped with comfortable cabs these difficulties have been largely solved; however, the stresses of vibration and noise inherent in the use of tractors remain.

Through a survey conducted by Sara et al. [4] appears that the primary motivations for investing in agricultural robots are to reduce labor costs while the availability of economic incentives and environmental sustainability appear to be less important factors. Several authors stated that the use of robotic technology can contribute to achieving several of the United Nations Sustainable Development Goals (SDGs). The smaller size of autonomous machines, compared to conventional tractors and equipment, helps reduce soil-related problems, particularly soil erosion and compaction, which are often caused by larger and heavier machinery [11]. In fact, it is well known that agricultural robots should meet specific criteria, such as being lightweight, compact, autonomous, intelligent, communicative, safe, and adaptable, to perform tasks efficiently [147–149]. Moreover, the adoption of precision agriculture procedures to optimize the use of resources and increase the timeliness of crop operations through, for example, direct seeding, mechanical weeding, or ultra-low volume spraying, allows farmers to produce more with less. But, as the proposed study points out, it should certainly be considered that a major limitation is the fact that most robots are designed to automate specific tasks. For this reason, the purchase of poorly adaptable robots is considered a rather high risk when humans are instead able to switch between tasks. Other aspect may raise concerns in relation to social sustainability, although in advanced economies the number of workers in agriculture is declining and, in any case, the demand for skilled labor is not being met. At the same time, the reduction of drudgery can improve the livelihoods of farmers, especially small-scale farmers [4].

In conclusion, the review made it possible to highlight that companies, entrepreneurs and research groups are devoting more and more interest to robotics because of the many benefits associated with its use in agriculture from different point of view. Events dedicated to agricultural robotics have been organized for a number of years (such as the World FIRA), and it is not uncommon to find specimens and prototypes at the main international events concerning agricultural mechanization, such as Agritechnica, FIMA and EIMA International. This demonstrates the growing interest in these technologies by farmers and manufacturers especially from younger generations that are attracted to digital technology, particularly ground and air robots. However, in summary still other challenges need to be addressed, such as the development of a stable and reliable method for environmental mapping and route planning, which is of great importance for the autonomous function of agricultural robots in orchards. But also the human-robot and machine-robot collaboration represents a promising challenge for some agricultural activities as well as other efforts should be directed at evaluating the energy performance of robotic platforms, the batteries efficiency and the energy supply in open field [150].

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